MIL-T-50782(MU)
25 August 1972

MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M117A2

1. SCOPE

1.1 This specification covers the Panoramic Telescope, M117A2 which employs a counter mechanism.

2. APPLICABLE DOCUMENTS

2.1 The following documents. of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

ML-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

Fl1739510	Telescope, Panoramic: M117A2

PACKAGING DATA SHEET

11739510	Packaging of Telescope, Panoramic: M117A2

FSC 1240

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The telescope shall be manufactured In accordance with Drawing F11739510 and drawings pertaining thereto.

3.2 General specification. - The contractor shall be responsible for adherence to and compliance with the requirements of Specification MIL-F-13926 specified below. Should any conflict exist between the general specification and this specification, this specification shall take precedence.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification and marking.
(f) Workmanship.

3.3 Environmental.

3.3.1 Storage temperature. - The telescope shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature (73 degrees ± 18 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.3.2 Operating temperature. - The telescope shall meet the applicable requirements of 3.8.2.6 to 3.9.2.3 inclusive while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit. Upon return to standard ambient temperature, the telescope shall meet the requirements of 3.3.3.

3.3.3 Vibration. - The telescope shall be vibrated in a vertical plane at a frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of physical failure and the telescope shall meet the requirements of 3.4 to 3.12 inclusive. (see 3.7.1.1)

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3.4 Sealing.

3.4.1 Pressure (optical system). - The entire interior area of the optical assembly shall be capable of withstanding an internal pressure of 5 ± 0.10 pounds per square inch (psi) for a period of 5 minutes with no evidence of leakage.

3.4.2 Pressure (counterbox area). - Apply an internal, pressure of 2.0 + .25 pounds per square inch (psi) of dry nitrogen for a period of 3 minutes to each telescope. These shall be no evidence of leakage.

3.5 Cleanliness.

3.5.1 Dirt. - Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be three (3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.5.2 Other defects. - There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations when viewing through the objective and eye end of the instrument.

3.6 Orientation. - The requirements of 3.7 to 3.8,3.3.2 inclusive shall be met with the telescope oriented in the following position:

(a) The main telescope tube vertical as determined by the keys on each side.
(b) The reset and 6400 mil counters indicating 3200 mils
(c) The gunner's aid counters set at zero.
(d) The line of sight horizontal.
(e) The elbow assembly centered between the swing stops within 2°.
(f) Knob 8587442 indicating INDIRECT (out of detent).

3.7 Accuracy.

3.7.1 Collimation.

3.7.1.1 Pre-vibration. - Prior to meeting the requirements of 3.3.3, the line of sight shall be perpendicular to the locating key-ways within 5.0 mils

3.7.1.2 Post-vibration. - Subsequent to meeting the requirements of 3.3.3, the line of sight shall remain within 0.25 mil of the azimuth position established in 3.7.1.1.

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3.7.2 Image tilt. - When checked at 1600 mil increments in azimuth through two full revolutions in each direction, the image of a vertical target line shall be vertical within two degrees of arc. This requirement shall also be met with the elbow assembly positioned against each of the swing stops.

3.7.3 Parallelism of reticle and image. - The vertical reticle line shall be parallel to the image of a vertical target line within 30 minutes of arc at the position established in paragraph 3.6.

3.7.4 Eyepiece focus. - The eyepiece focus of the reticle image shall be fixed between the limits of -0.50 and -1.0 diopter.

3.7.5 Patallax. - Parallax in the center of the field shall not exceed 0.1 mil when viewing a target at 80 meters ± 1.0 meter.

3.7.6 Resolution. - Resolution on the optical axis shall be 10.0 seconds of arc or less when using an auxiliary telescope of at least 3 power having an entrance pupil of at least 0.4 inch. Resolution shall consist of detectable line structure and proper line count in all four meridians. Total spread in focus shift between meridians shall not exceed 0.25 diopters as measured at the telescope eyepiece.

3.7.7 Reticle accuracy. - The angular separation between the two 40 mil graduations on the horizontal axis shall be 80 mils plus or minus 0.5 mil.

3.7.8 Image wander. - The wander of the reticle image measured at the eyepiece shall not exceed two degrees in diameter for two (2) complete revolutions of the head assembly. In addition, all reticle graduations shall be visible throughout the revolutions of the head. This requirement shall also be met with the elbow assembly positioned against each of the swing stops.

3.8 Performance.

3.8.1 Elevation mechanism.

3.8.1.1 Excursion range. - Rotation of the elevation knob shall cause the line of sight to elevate and depress a minimum of 300 mils from the position specified in 3.6(d).

3.8.1.2 Plumb travel. - Starting with the line of sight in coincidence with a vertical target line at zero elevation, the line of sight shall not deviate by more than 0.5 mil (total spread 1.0 mil) under the following conditions:

(a) When the line of sight is elevated 300 mils from zero elevation.

(b) When the line of sight is depressed 300 mils from zero elevation.

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3.8.1.3 Side Play. - Reversing movement of the elevation knob at zero elevation shall not cause the line of sight to shift more than 0.2 mil in azimuth.

3.8.2 Azimuth mechanism.

3.8.2.1 Azimuth error. - Error at l zimuth readings, taking in 800 mil increments through two full revolutions shall not exceee 1.0 mil excluding backlash.

3.8.2.2 Deflection error. - Rotating the gunnner's l aid knob to indicate 15 mils followed by rotation of the azimuth knob in the opposite direction (with backlash removed) to deflect the line of sight 15 mils shall cause the reset counter to return to within 0.3 mil of the original setting. This requirement shall be met with the 15 mil value set into each scale and the azimuth counter indicating 3200, 4800 zero, and 1600 mils at the start of each reading .

3.8.2.3 Backlash. - Backlash in the azimuth mechanism shall not exceed l.O mil when read on the azimuth counter; 1.5 mils when read on the reset counter.

3.8.2.4 Level travel. - Starting with the line of sight in coincidence with a horizontal target, the line of sight shall track the target within 6.0 mils (total excursion) when rotated through two full revolutions.

3.6.2.5 Lift. - Vertical displacement of the line of sight due to reversing rotation by means of the azimuth knob shall not exceed 2 mils.

3.8.2.6 Five mil click lead mechanism. - positioning Knob 8587442 to indicate DIRECT shall cause a detent action to be felt when introducing 5.0 ± 0.5 mil increment changes into the 6400 mil counter. This requirement shall be met at standard and extreme operating temperatures.

3.8.3 Counter mechanism.

3.8.3.1 Rapid traverse. - The reset counter shall follow the azimuth counter within 0.25 mil when the knob is rotated at 200 rpm for one minute. The gunner's aid counters shall remain at zero.

3.8.3.2 Reset counter. - The reset counter shall be resettable to 3200 ± 0.25 mils from any reading.

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3.8.3.3 Gunner's aid counters.

3.8.3.3.1 Excursion range. - The gunner's aid counters shall have a minimum of 50 mils excursion on the "L" and "R" scales.

3.8.3.3.2 Settings. - Setting 50 mils on the "L" scale shall decrease the reset counter reading by 50 ± 0.5 mils. Setting 50 mils on the "R" scale shall increase the reset counter reading by 50 ± 0.5 mils There shall be no change in the azimuth counter readings when the 50 mils values are set into the gunner's aid counters.

3.9 Operability. - The boresight adjustment shaft, rheostat knob, azimuth counter door and latch, positioning of crank and positioning of Knob 8587442 shall be operable at standard and extreme operating temperatures.

3.9.1 Torque. - The running torque required to operate the following knobs shall be within the values specified for standard temperature and shall not exceed the values specified for extreme operating temperatures. The torque values for the azimuth knob shall be met with Knob 8587442 indicating INDIRECT (out of detent).

Knob	78 and 18 F (Standard Temperature)	-40° and +150°F (Extreme Operating Temperatures)
Azimuth	2 to 6 in. - lbs.	9 in. - lb.
Gunner's aid	10 to 36 in. - ozs.	60 in. - ozs.
Reset Counter	10 to 36 in. - ozs.	46 in. - ozs.
Elevation	10 to 48 in. - ozs.	86 in. - 0zs.

3.9.2 Direct force. - The following operability requirements shall be met with the force applied horizontally and in the direction of movement.

3.9.2.1 Elbow assembly. - When checked at standard temperature, a direct force of 2 to 7 pounds applied adjacent to the eyelens shall be required to rotate the elbow assembly in each direction through the full extent of travel. The force required at extreme operating temp eratures shall not exceed 13 pounds.

3.9.2.2 Reset counter knob. - When checked at standard temperature, a direct force of 7 to 12 pounds shall be required to push the reset

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counter knob in sufficiently to permit rotation. The force required at extreme operating temperatures shall not exceed 15 pounds.

3.9.2.3 Count numerals. - With the lowest digit of each counter in coincidence with its respective index line, the remaining numbers shall be in horizontal alignment within one-sixteenth inch.

3.10 Illumination.

3.10.1 Counters. - With the lamp energized, the numerals and markings of each shall be clearly distinguishable when observed in a darkened area.

3.10.2 Reticle. - With the reticle lamp energized, the reticle markings shall be visible when observed in a darkened area. This requirement shall be met through one full revolution of the head in azimuth. Rotation of the reticle rheostat knob shall vary the intensity of illumination.

3.11 Insulation resistance. - Insulation resistance shall not be less than 10 megohms when a potential of 100 vdc is applied.

3.12 Continuity. - The continuity of the wiring shall be checked for conformance with applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All All " parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing. - When the Government is responsible for conducting first article approval test, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the samples including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table I herein. Examination and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: 100% Inspection		
1. Pressure (optical assembly)	3.4.1	4.6.1.2
2. Pressure (counter assembly)	3.4.2	4.6.1.2.1
MAJOR: AQL 1.0 Defective		
101. Pre-vibration collimation	3.7.1.1	4.7.2.1
102. Vibration	3.3.3	4.7.2.1.1
103. Post vibration collimation	3.7.1.2	4.7.2.1.2
104. Cleanliness	3.5	4.7.2.1.3
105. Image tilt	3.7.2	4.7.2.1.4

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CHRACTERISTIC	REQUIREMENT	TEST PROCEDURE
106. Parallism of image and reticle	3.7.3	4.7.2.1.5
107. Eyepiece focus	3.7.4	4.7.2.1.6
108. Parallax	3.7.5	4.7.2.1.7
109. Reticle accuracy	3.7.7	4.7.2.1.8
110. Image wander	3.7.8	4.7.2.4
111. Settings counter gunner's aid	3.8.3.3.2	4.7.2.2.1
112. Counter excursion range	3.8.3.3.1	4.7.2.2.1
113. Azimuth error (800 roils)	3.8.2.1	4.7.2.2.2
114. Azimuth deflection error (15 mils)	3.8.2.2	4.7.2.2.2
115. Azimuth backlash	3.8.2.3	4.7.2.2.2
116. Level travel	3.8.2.4	4.7.2.2.3
117. Lift	3.8.2.5	4.7.2.2.4
118. Click mechanism	3.8.2.6	4.7.2.2.5
119. Rapid traverse (reset counter)	3.8.3.1	4.7.2.2.6
120. Reset counter (reset operation)	3.8.3.2	4.7.2.2.6
121. Elevation travel	3.8.1.1	4.7.2.3.1
122. Plumb travel	3.8.1.2	4.7.2.3.1
123. Side play	3.8.1.3	4.7.2.3.1
124. Resolution	3.7.6	4.7.2.2.7
125. Torque (73°±18°F)	3.9.1	4.7.2.4.1
126. Operability (73°±18°F)	3.9	4.7.2.5
127. Illumination (reticle)	3.10.2	4.7.2.6
128. Illumination (counters)	3.10.1	4.7.2.6
129. Counter number alignment	3.9.2.3	4.7.2.7
130. Elbow movement force (73°18°F)	3.9.2.1	4.7.2.8
131. Reset counter force (73°±18°F)	3.9.2.2	4.7.2.9
132. Insulation resistance	3.11	4.7.2.10
133. Continuity	3.12	4.7.2.11

4.3.3 Acceptance and reflection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling for environmental testing.

4.4.1 Samples shall be selected form lots at random using inspection level S-3 Table I of MLL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and test. The requirements outlined in 3. 1 and 3.2 shall be accepted on contractor's certification.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Temperature (storage)	3.3.1	4.6.1.1
302. Temperature (operating)	3.3.2	4.6.1.1
303. Operability (operating temperature)	3.9	4.7.2.5

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CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304. Torque (operating temperature)	3.9.1	4.7.2.5.1
305. Elbow movement force (operating temperature)	3.9.2.1	4.7.2.8
306. Reset counter force (operating temperature)	3.9.2.2	4.7.2.9

4.4.2 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract Provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Govewrnment design. - Unless otherwise specified in the contaract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.05.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment

4.5.3 Test equipment. - In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and 4.7.

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NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926
2. Continuity checker	Standard shop type capable of providing readings accurately within 2% of specified tolerance
3. Insulation resistance checker	Standard shop type conforming to the accuracies outlined in 4.5.2.2
4. Power supply	0 to 30 vdc output within 10% accuracy
5. Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature test. - The panoramic telescope shall be subjected to test Procedure I of MIL-F-13926; however, the temperatures specified in 3.3.1 and 3.3.2 shall apply. When the panoramic telescope has been stabilized at the temperature specified, the result of the specified test shall conform to the limits outlined in 3.3.1 and 3.3.2 using test procedures outlined herein.

4.6.1.2 Sealing optical assembly. - This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy depicted on Drawing F8565556. When using pressure test fixture, the fixture shall be set-up as outlined in set-up instructions shown on Drawing F8565556, Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the panoramic telescope, adjust testing pressure to the pressure specified in 3.4.1 and maintain the pressure for time specified in 3.4.1. Pressure gage shall remain constant to assure compliance with 3.4.1. At the completion of this test, connect a dew point tester to the exit post and purge the optical assembly until -25°F is recorded and seal the instrument.

4.6.1.2.1 Sealing (counter assembly). - This test to be accomplished as outlined above, however, the pressure and time values specified in 3.4.2 shall apply.

4.7 Performance testing.

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4.7.1 Orientation. - Proper orientation of the panoramic telescope is assured when mounted and secured to an adapter, the accuracy of which shall be equal to or better than the accuracy depicted on Drawing F8599913, and when aligned as outlined in set-up instructions specified on Drawing F8599913. With the panoramic telescope positioned on the the fixture as outlined above the line of sight, elbow assembly, counters and the click mechanism shall be set as specified in 3.6.

4.7.2 Collimation.

4.7.2.1 Pre-vibration. - With telescope positioned as outlined in 4.7.1, observe coincidence between the panoramic telescope line of sight and the optical axis of the test reticle (target) in the collimator projector. This coincidence must be within the limits outlined in 3.7.1.1. Record this position. Exercise care not to move the setting of the panoramic telescope.

4.7.2.1.1 Vibration. - This test shall be performed using a testing device, the accuracy of which shall be equal to or exceed the accuracy depicted on Fixture Drawing F7560085. The test shall be performed as outlined in Procedure V of MIL-F-13926; however. the duration shall be as specified in 3.3.3. At the conclusion of the test, the telescope shall be subjected to a visual and tactile examination and shall be subjected to the following performance test to determine conformance with 3.3.3

4.7.2.1.2 Post vibration. - Immediately after completing the vibration test, re-assemble the telescope to the adapter and without disturbing any preoius settings, observe where the panoramic telescope's optical line of sight falls upon the projector collimator reticle. This observed point and the recorded position (4.7.2.1) shall coincide within the limits specified in 3.7.1.2.

4.7.2.1.3 Cleanliness. - The panoramic telescope shall be tested for cleanliness by using the shadowing technique outlined in MIL-O-13830. At the conclusion of this test, dirt particles shall not exceed the limits outlined in 3.5.

4.7.2.1.4 Image tilt. - With the panoramic telescope and fixture positioned as outlined in 4.7.1, image tilt shall be observed by sighting through Collimator F5549108 aligned to the optical axis of the eyepiece assembly and note parallelism between its vertical reticle line and the vertical line of the collimator target. Both lines shall be parallel within the limits provided on the projected target (collimator reticle) to assure conformance with the 3.7.2. Repeat this test With the elbow assembly against each of the stops to assure conformance with 3.7.2. Again observe parallelism between the reticle lines. Repeat this test at eight different azimuth counter settings in 6400 roils of rotation. Each setting in azimuth will require the azimuth fixture plate to be rotated an equal amount in an opposite direction to permit

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re-establishing the line of sight to the position observed in 4.7.2.1.2. All readings are to be within the limits outlined in 3.7.2.

4.7.2.1.5 Parallelism of image and reticle. - With the panoramic telescope positioned as outlined in 4.7.1, align one end of the instruments vertical reticle line to the projector collimator's vertical reticle line and observe where the opposite end of the panoramic telescope vertical reticle line falls with respect to the projector collimator vertical reticle line. Both lines must conincide wihtin the limits specified in 3.7.3.

4.7.2.1.6 Eyepiece focus. - This test is to be accomplished by placing a white background in front of the panoramic telescope window. Eyepiece focus is to be measured using a 3 power dioptometer conforming to the accuracies depicted on Drawing 7680631. Adjust the dioptometer eye-piece assembly to the dioptometer's reticle until a clear sharp image appears on the dioptometer's optical axis. Next direct the dioptometer's line of sight toward a 500 feet distant target such as a smoke stack, edge of a building, or some other well defined target image. Adjust the objective assembly until parallax is removed and set the diopter scaloe to zero. Parallax is defined as any apparent movement between the reticle and target image when the head is moved from side to side or up and down. Align the dioptometer's optical axis to the panoramic telescope optical axis and obtain a clear sharp image of the panoramic telescope reticle on the optical axis. Make all measurements on the dioptometer by adjusting the objective assembly. Any and all adjustments to be within the limits of 3.7.4.

4.7.2.1.7 Parallax. - Using the set-up outlined in 4.7.1, parallax will be tested after the distance specified in 3.7.5 has been set into the projector colimator. The test is performed by aligning a reticle line or mark near or on the optical axis of the telescope under test to a similar line or mark on the projector collimator reticle. There shall be no apparent movement beyond the limits of 3.7.5 between the selected lines or mark when the observer moves his head from side to side or up and down. Reticle displacement shall not exceed the limits outlined in 3.7.5. Use the thickness of the reticle line to measure apparent movement.

4.7.2.1.8 Reticle accuracy. - Angular separation between the two 40 mil graduations may be determined by aligning one of the panoramic telescope's 40 mil graduation to the corresponding mark of the projected reticle or wall targert. This alignment may be attained by activating the elevation and azimuth knobs. Observe coincidence between the other 40 mil graduation of the panoramic telescope and the corresponding mark

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of the projected reticle or wall target. This coicidence must be within the limits of 3.7.7.

4.7.2.2 Counter operation.

4.7.2.2.1 Gunner aid counters. - This test may be accomplished with the panoramic telescope positioned as outlined in 4.7.1 with the line of sight falling upon the projector collimator's reticle as outlined in 4.7.2.1.2. Observe and record the azimuth test fixture reading. Set any correction reading into either the "L" or "R" counter. Any reading induced into the right "R" gunner's aid counter shall increase the reset counter an equal amount (±0.5) and any reading induced into the left "L" gunner's aid counter shall decrease the reset counter an equal amount (±0.5) to assure compliance with 3.8.3.3.2. When 50 mils has been inserted into either gunner's aid counter, the reset counter shall reflect an increase or decrease of 50 roils within the limit outlined in 3.8.3.3.2 and the correction counter shall have traveled the minimum excursion outlined in 3.8.3.3.1 and the 6400 mil counter shall remain unchanged (see 3.8.3.3.2). The test for accuracy shall be performed by inserting a correction into either the "L" or "R" gunner's aid counter and observing the appropriate change in the reset counter. Turn the azimuth fixture plate to traverse the line of sight in a clockwise direction an equal amount when a correction is placed into the "L" counter. By means of the azimuth knob of the panoramic telescope traverse the line of sight in the opposite direction until the reset counter records 3200 mils. Not sighting through the panoramic telescope observe coincidence between the panoramic telescope's optical axis and the point on the projector collimator's reticle established in 4.7.2.1.2. Repeat this procedure at two additional correction readings and at the 50 mil correction setting of each gunner's aid counter.

4.7.2.2.2 Azimuth error, backlash and deflection error. - With the panoramic telescope positioned as outlined in 4.7.2.1, sight through the panoramic telescope and by means of the azimuth and elevation knobs coincide the telescope's line of sight (intersecting point of the horizontal and vertical center reticle lines) to the geometric axis of the collimator projector (intersecting point of the horizontal and vertical center reticle lines).Observe that the azimuth and reset counters read 3200 mils and both gunners aid counters read 0 mils. Set the azimuth fixture off 800 roils in one direction and offset the panoramic telescope's line of sight in the opposite direction until coincidence previously mentioned has been obtained. Do not override this point of coincidence. Record the reset and azimuth readings. Continue rotating the azimuth knob (panoramic telescope) beyond this point of coincidence one revolution and return to the previous point of coincidence. bserve reset and azimuth counter readings. The azimuth counter and the fixture reading must coincide within the limits of 3.8.2.1. The difference between this reading and the azimuth counter reading previously recorded must not exceed the limits of 3.8.2.3. The difference between this reset counter reading and the previously

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recorded reset counter reading must not exceed the limits of 3.8.2.3. Repeat this procedure at the azimuth settings specified in 3.8.2.1 and also from the opposite direction. Insert 15 mils ddflection into the azimuth fixture and insert 15 mils correction into the appropriate correction counter. The reset counter shall reflect this 15 mil correction. By means of the azimuth knob (panoramic telescope) return the line of sight to the previously recorded point of coincidence and observe the reset counter reading. This reading must coincide with the fixture setting within the limits of 3.8.2.2. Repate this procedure at the azimuth settings specified in 3.8.2.2 and also using the other correction counter.

4.7.2.2.3 Level travel. - This test may be performed in conjunction with the azimuth test outlined in 4.7.2.2.2. The test is performed by sighting through the panoramic telescope after a selected azimuth setting has been inserted into the fixture and panoramic telescope, observe coincidence between the telescope horizontal reticle line and the horizontal target line near the optical axis. They must coincide with the limits outlined in 3.8.2.4.

4.7.2.2.4 Lift. - This test may be performed in conjunction with the backlash test outlined in 4.7.2.2.2. The test is performed after returning the line of sight to the point of coincidence by observing where the horizontal reticle line of the panoramic telescope falls with respect to the previously established position (4.7.2.2.3). They must coincide within the limits of 3.8.2.5.

4.7.2.2.5 Five mil click mechanism. - This test may be performed by using visual and tactile inspection techniques. The result of this inspection shall conform to the requirement outlined in 3.8.2.6 when performed at the temperature specified.

4.7.2.2.6 Rapid traverse (reset counter). - This test is made by assembling a testing mechanism capable of providing an RPM output specified in 3.8.3.1 to the azimuth knob of the panoramic telescope by means of an appropriate adapter. While being driven at the RPM for the time specified, the reset counter shall follow the azimuth counter as required in 3.8.3.1. At the end of the speed test, the reset counter shall reset to 3200 mils as specified in 3.8.3.2.

4.7.2.2.7 Resolution. - This test may be performed after directing the line of sight toward a target similar to that specified in MIL-O-13830. The actual test is performed as outlined in MIL-O-13830. Observe the definition pattern on the target. The patterns must be resolved in the four meridians within the limits outlined in 3.7.6.

4.7.2.3 Elevation.

4.7.2.3.1 Elevation and plumb travel side play. - Return the Panoramic telescope to the position outlined in 4.7.1. Scribe an index line on

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the elevation knob in line with another index scribed on Housing 11739512. Perform these tests by rotating the azimuth fixture plate to direct the panoramic telescope's optical axis toward a plumb line located to one side of the fixture and coincide the center vertical reticle line to the plumb line. Position the vertical target line in coincidence with the plumb and vertical reticle line and raise or lower the target to obtain coincidence between the target horizontal line and the horizontal reticle line of the panoramic telescope's reticle. The plumb travel and side play test are performed by actuating the panoramic telescope's elevation knob from the zero setting to each extreme. During the plumb travel test the panoramic's line of sight must track the plumb line within the limits line of the target to assure conformance with 3.8.1. Return the line of sight to zero elevation and observe coincidence between the panoramic's optical axis and the original point of coincidence. Any non-coincidence shall not exceed the limits specified in 3.8.1. Use the width of the reticle line to make this measurement. Per for this procedure moving the panoramic's line of sight to the other elevation extreme.

4.7.2.4 Image wander. - With the panoramic and fixture positioned as outlined in 4.7.1 align the auxiliary telescopes optical axis to the panoramic's optical axis. With these points in coincidence by means of the azimuth knob rotate the head two full revolutions. The The coincided points must not spread more than the limit specified in 3.7.8. Use the two degree circle of the auxiliary telescope's reticle when measuring image wander.

4.7.2.5 Operability. - When the panoramic telescope has been stabilized at the temperatures specified in 3.9, tactile inspection techniques shall be used to determine compliance outlined in 3.9.

4.7.2.5.1 Torque. - Torque tests shall be performed under temperature conditions outlined in 3.9.1. The torque tests shall be performed using a torque wrench and appropriate adapter. The torque readings obtained shall conform to the limits specified in 3.9.1.

4.7.2.6 Illumination. - Electrically connect a power source to the panoramic telescope capable of producing 24 ± 6 vdc and energize the appropriate lamps and visually inspect to determine compliance with 3.10.1 and 3.10.2. Rotation of the rheostat knob from the off position shall increase intensity of illumination from dim to bright.

4.7.2.7 Counter numbers. - By means of standard measuring equipment determine digit alignment in accordance with 3.9.2.3.

4.7.2.8 Force applied to elbow assembly. - Secure an appropriate adapter to the elbow assembly at the eyepiece assembly and apply a push force. The force required to move the elbow assembly through full travel shall not exceed the limits outlined in 3.9.2.1 for the temperatures specified.

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4.7.2.9 Restet counter force. - Secure an appropriate adapter to the reset counter and apply a pushing force to disengage the locking mechanism of the reset counter. This force will be measured when the reset knob is permitted to rotate and shall not exceed the limits of 3.9.2.2.

4.7.2.10 Insulation resistance. - This test shall be performed as outlined in Test Method 302 of MIL-STD-202, using test condition "A". The test shall be made between insulated points and ground. Resistance shall not be less than the value specified in 3.11.

4.7.2.11 Continuity. - Use standard shop type continuity checks in conjunction with standard inspection techniques to determine conformance with 3.12.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet 11739510. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Panoramic Telescope, M117A2 when mounted in Telescope Mount, M145, is primarily intended to lay the Self-Propelled Howitzer 155MM, M109A1 in azimuth for indirect fire.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet number (see 5.1)

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A388

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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